DASARI BALA SRINIVAS

Mechanical Engineer | CAD & CAE | Systems Engineering, Design & Integration | Defence, Aerospace & Energy
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SUMMARY

Mechanical Engineer with a first-principles approach to problem-solving, developing expertise through hands-on experience in deep-tech and R&D Startup Environments. Experienced in mechanical design, CAD/CAE, electronics packaging, and multidisciplinary engineering, with an emphasis on understanding system-level requirements, managing interfaces, and integrating subsystems into practical engineering solutions. Driven by continuous learning, technical ownership, and solving complex engineering challenges.

SKILLS & TECHNICAL CAPABILITIES

- **CAD & CAE:** SolidWorks, CATIA, ANSYS and SimScale.
- **Mechanical Systems Design:** Concept development, requirement-to-design translation, custom chassis and enclosure design, precision mechanisms, mounting interfaces, structural assemblies and design optimisation.
- **Defence & Aerospace Electronics Packaging:** radar-system mechanical design and RF subsystem integration, VPX-based electronics packaging, Line Replaceable Unit design and integration, test and qualification chassis, air-cooled, conduction-cooled, and liquid-cooled chassis design, MIL-STD, VITA standards and applicable Indian Standards (IS).
- **Engineering Documentation:** Interface Control Documents (ICDs), Quality Assurance Plans (QAPs), Bills of Materials (BOMs), Compliance Documents, SOW, CAE reports, engineering drawings, Pre-visualisation and CAD rendering.

EXPERIENCE

Avarthana Technologies Pvt. Ltd. *Bengaluru, Karnataka, India*…….. 10 Months
Research and Development Engineer……. July 2026 - Present (4 months)
Research and Development Intern……... January 2026 - July 2026 (6 months)

- Supported R&D activities in radar development, VPX systems, and custom MIL-STD applications through CAD/CAE-driven design and engineering analysis.
- Collaborated with multidisciplinary RF, FPGA, and Civil Engineering teams during the pre-development of a specialised radar system, contributing to mechanical structures, thermal management, interfaces, mounting arrangements, and pre-visualisation models.
- Contributed to new designs and redesigns of LRUs, test and qualification chassis, air-cooled, conduction-cooled, and liquid-cooled systems, and mechanical assemblies. Conducted self-directed R&D into precision mechanisms, particularly flexures.

Away Space Covenant (AwSc). *Guntur, Andhra Pradesh, India*…… August 2025 (1 month)
CAD & CFD Engineering Intern……. August 2025 (1 month)

- Contributed to the design and development of an Earth-observation product by creating prototype models in CATIA and performing CFD simulations in ANSYS Discovery and Fluent to evaluate and optimise aerodynamic performance. Collaborated with the engineering team to develop a functional prototype, gaining practical experience in research-driven product development and aerospace engineering.

PROJECTS

Battery Thermal Management System for Electric Two-Wheelers | Capstone Project…….. **April 2026**
Designed a hybrid PCM–EG and heat-pipe battery thermal management system for a 200-cell electric two-wheeler battery pack. Developed the mechanical assembly in SolidWorks and performed transient thermal simulations in SimScale. Achieved a 6°C reduction in simulated peak temperature at 40°C ambient, while demonstrating predictable cyclic thermal behaviour under the hybrid configuration and identifying rapid thermal escalation toward potential thermal runaway in the PCM–EG-only case.

Atmospheric Water Generator …….. **April 2022**
Designed and fabricated an atmospheric water generator prototype capable of producing up to 10 L/day through dew-point cooling. Integrated an R134a-based vapour-compression refrigeration cycle using repurposed air-conditioning and refrigeration components to condense atmospheric moisture. Presented the prototype at Regional Polytech Fest 2022, receiving appreciation for its innovative approach and resource-efficient component reuse.

EDUCATION

Bachelor of Technology - Mechanical Engineering …….. **CGPA: 7.5**
Vasireddy Venkatadri Institute of Technology, Nambur, Guntur……... August 2023 - June 2026

Diploma in Mechanical Engineering…….. **Percentage: 73.44%**
M.B.T.S Government Polytechnic, Nallapadu, Guntur…….. October 2020 - April 2023

High School, *St Ann's EM School*…… March 2020, **Percentage: 96%**

CERTIFICATIONS

- **Master CAD/CAM - Central Institute for Tool Design**…….. December 2022 - June 2023 (6 months)
- **Engineering Design and Simulation - Professional Certificate - MathWorks**…….. edX, September 2024
- **NPTEL - Computer Integrated Manufacturing - (Elite)**…….. January - April 2024
- **CATIA Expert Surface Design – APSSDC**…….. May – July 2025